

Weekly report: High-pressure and High-temperature Project

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Question

Why is ODMR measurement unstable or inefficient under blue-laser excitation?

Hypothesis

A short-wavelength laser has a high photon energy, so it can efficiently induce photoionization of NV^- : $\text{NV}^- \rightarrow \text{NV}^0 + e^-$

Evidence

(1) ODMR contrast: C_{ODMR} depends on the NV^- population: $C_{\text{ODMR}} = \frac{P_- C_- F_-^{\text{off}}}{P_- F_-^{\text{off}} + (1 - P_-) F_0 + B}$ where P_- is the NV^- population, C_- is the intrinsic ODMR Contrast of NV^- , F_0 is NV^0 fluorescence, and B is background

(2) The rate-equation simulation reproduces previous experimental trends(図 1)

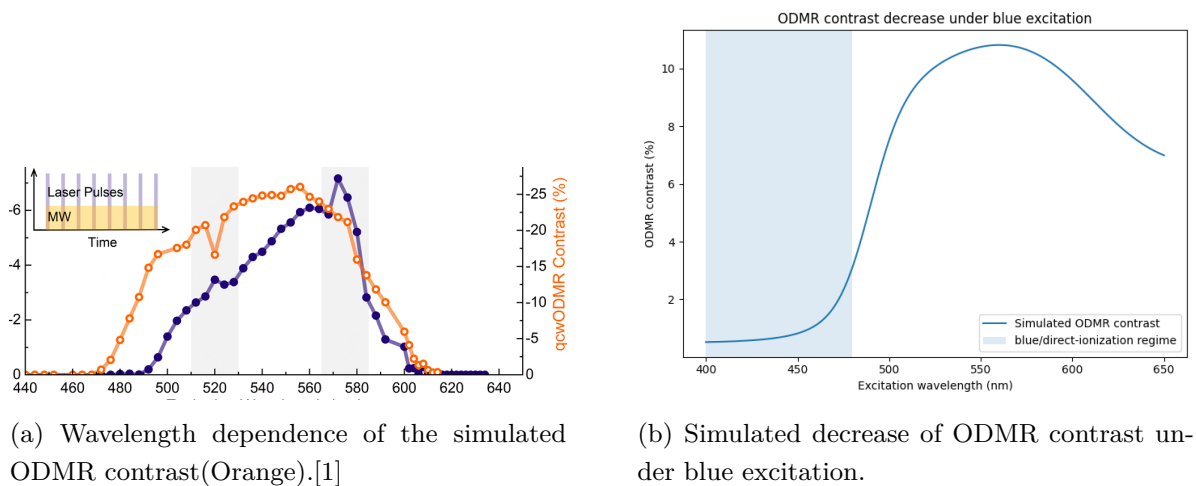


図 1: Simulation of ODMR contrast reduction caused by photoionization from NV^- to NV^0 under short-wavelength excitation.

Gap/Failure

- (1) The current model is phenomenological. It explains the broad trend that blue excitation reduces ODMR contrast, but it does not fully determine microscopic ionization pathways.
- (2) The current simulation may overestimate the charge stability at long wavelengths

Next step

The next step is to extend the model to include two-photon excitation and two-step charge conversion.

Reference

- [1] Lina M. Todenhagen, et al., arXiv:2307.11830